# Science

Nature Walks & Scouting
General Science
Labs
Natural History
Nature Notebook

Science: Grade 6

GRADE(S): 6

## About the Course

In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level, historical context, and time required are transitional as we gently prepare for the middle grades.

This course includes the following topic(s): General Science: Grade 6, Natural History: Grade 6, Labs: Grade 6, Nature Notebook: Grade 6, Nature Walks & Scouting: Grades 1-8

About Nature Walks & Scouting: Grades 1-8
Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About General Science: Grade 6
Learners are introduced to scientific activities and technology with a nod toward historical context. Specifically, students learn about plate tectonics, anatomy and medicine, and architectural engineering. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

About Labs: Grade 6
This is the required lab component for level 6 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

About Natural History: Grade 6
Students learn about oceanography, predator-prey relationships, and the impact of pollution as they consider man's stewardship of the natural world. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 6
Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 6
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

## Placement & Combining Tips

Science: Grade 6
For sixth-grade students or possibly hungry fifth-graders or seventh-graders taking their time.  If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, learners may stay at their appropriate level for General Science while combining Natural History and Outdoor Work, or teachers can adapt the laboratory content and reading level to meet their needs.

Nature Walks & Scouting: Grades 1-8
Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

General Science: Grade 6
For sixth-grade students or possibly hungry fifth-graders or seventh-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Labs: Grade 6
The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.

Natural History: Grade 6
For approximately sixth-grade students based on a balance of complexity and reading level. However, learners may be freely combined, or sequence reordered based on needs and preferences.

Nature Notebook: Grade 6
Learners may be combined and share prompts or follow their own interests. Support through varied media, scribing observations, or notebooking in tandem.

## Scheduling

| GRADE | SCHEDULE INFO. | BOOKS |
|---|---|---|
| 1-8 | Nature Walks & Scouting: Grades 1-8<br>1 time/week 30 min+ | |
| 6 | General Science: Grade 6<br>2 times/week 30 min | Plate Tectonics<br>The Art of Construction<br>Phineas Gage |
| 6 | Labs: Grade 6<br>1 time/week 30 min | Science: Grade 6 Lab Book |
| 6 | Natural History: Grade 6<br>1 time/week 15 min | The Frog Scientist<br>The Wolves and Moose of Isle Royale: Restoring an Island Ecosystem<br>Tracking Trash |
| 6 | Nature Notebook: Grade 6<br>1+ time/week 15 min+ | |

### Sample Weekly View

| Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|---|---|---|---|---|
| **Science: Grade 6** | | | | |
| Natural History: Grade 6 | | General Science: Grade 6<br>Nature Walks & Scouting: Grades 1-8 | General Science: Grade 6<br>Nature Notebook: Grade 6 | Labs: Grade 6 |

## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 6
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Labs: Grade 6
☐ Preview science labs and purchase materials, or have students gather them. Print or shortcut any links, as desired. While it does not usually matter when the week's lab is done in reference to the week's readings, this is not the case for Art of Construction in Term 3. The lab instructions for this book are embedded in the text, and the text is most living when they are completed alongside the reading. Therefore, it is best for lab day to fall on the third lesson of the week. If students' lab day is earlier in the week, teachers may want to delay the start of week 1, so that the third lesson is lab day. For example, if students have M/Th lesson days and T lab day, then students could begin lessons on Th of week 1. Alternatively, teachers may look ahead and adjust according to their preference.

### Special Topics & Field Trips

Science: Grade 6
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 6
☐ Term 1: any local geology site (e.g. a quarry or certain parks)
☐ Term 2: hospital open house
☐ Term 3: any buildings or bridges

Natural History: Grade 6
☐ Term 1: shore, stream, or reservoir, a marine animal rescue
☐ Term 2: wildlife conservationist
☐ Term 3: a pond (hopefully one where tadpoles may be collected)

## Term Prep & Teacher Tips

Science: Grade 6
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

General Science: Grade 6
☐ Term 1: specific rock types p.748-759
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• less than 1/2 c white vinegar
• recycled cardboard, roughly 3' x 3'
• books or wood blocks to adjust height
• 3 colored markers
• 2 or more apples (Term 2)
• plate
• knife
• cutting board
• plastic wrap
• roughly 2' x 2' piece of substrate (recycled cardboard, wood scrap, foam core)
• 15 sheets of multipurpose paper
• any weights for testing bridge
• various supplies by student design
• printables
• computer
• optional: camera, such as found on most electronic devices
• optional: antiseptic treatments
• optional: plastic ruler
• optional: fabric scrap
• optional: wax paper
• optional: fan/hair dryer
• optional: needle and thread
• optional: cereal box to model a building
• optional: rocks
• optional: sand paper
• optional: 3-7 weights with a hole, e.g. washers or spools
• optional: string or strong thread
• optional: hammer and nail
• optional: block of scrap wood
• optional: rope
• optional: nutcracker
• optional: 4 raw eggs (Term 3)
• optional: 8 sturdy egg cups
• optional: 8 cotton balls
• optional: scrap plywood (might be the same substrate as above)
• optional: cylindrical balloon
• optional: cup that fits the top of balloon
• optional: books of various sizes

Natural History: Grade 6
☐ Term 1: -
☐ Term 2: -
☐ Term 3: amphibians p.170-187

## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ [View Book List Details]

## Science: Grade 6

### General Science: Grade 6


Plate Tectonics


The Art of Construction


Phineas Gage

### Labs: Grade 6

Science: Grade 6 Lab Book

### Natural History: Grade 6


The Frog Scientist


The Wolves and Moose of Isle Royale: Restoring an Island Ecosystem


Tracking Trash

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# Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- [View Basic Supplies]
- [View Supply List Details]

### Nature Walks & Scouting: Grades 1-8


Wild Bird Seed


Bird Feeder


Hand Lens


Small Collection Containers


Small Basin


Dip Net


Student Microscope

### General Science: Grade 6

Alveary Grade 6 Science Kit

Household Items - General Science: Grade 6


Student Microscope


Prepared Microscope Slides

### Labs: Grade 6


Hand Lens


Scale/Balance


Classroom Collection of Rocks and Minerals

General Fossil Collection

## Quick Links

Click THIS text or scan the QR code for links.

### Science: Grade 6

- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android

### Labs: Grade 6

- Grade 6 Lab Book
- Lab Notebook Examples

# Science: Grade 6

## How To Teach

### Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

### Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

### Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

### Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

## Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

# Science: Grade 6

## How To Teach

### Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

• Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

“I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work.”

• Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

• After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

• If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.

### Lab Procedure

• Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

• The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

• If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

• Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.

## Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

• Similar to the prelab narration, the concluding narration is a chance to think about and put into words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

“It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn’t sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface.”

• Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

• Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

# Science: Grade 6

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## Term 1

**WEEK 1**

### ⬚ 15m Natural History: Grade 6 - Lesson 1

*Ocean Currents*

Materials: Tracking Trash

PREP: Read Teacher Tip.

**→ INTRO**
Look at the picture on the cover and the title. Have you ever noticed trash on the beach or in a stream? This book is about how one scientist uses that trash as an opportunity to learn more about the ocean and the creatures that live within it. Examine the map opposite p.1 and notice the current. Why do you suppose the mapmaker was interested in that current, and what does that have to do with our story?

**→ READ, NARRATE, & DISCUSS**
p.1-4 "Benjamin Franklin" - "learn from it."

• How do these ocean currents affect the climate? You can study the map on p.20-21 of your NatGeo World Atlas.
• Explain weather versus climate.

★ TEACHER TIP
If students read Tornado Scientist, they might recall that air masses are warmed by the Earth or water beneath them. Movement occurs based on their temperature and density, and this is a driving force behind weather patterns.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

**WEEK 1**

### ⬚ 30m General Science: Grade 6 - Lesson 1

*God Created*

Materials: The Holy Bible

**→ INTRO**
You have been learning about scientific activities in our modern world. Like all knowledge, our culture affects these activities, and in return, what we learn affects our culture. This has been going on throughout human history. Early scientists primarily studied the Things they could see and assumed that they were always this way, but eventually, they began to make some curious observations that suggested that maybe Things hadn't always been this way. These observations were shocking and confusing to these men and women. This was a major discovery that led many naturalists to question all they thought they knew and understood. It led some to reject science; others to reject their faith in God; and still others to be awestruck at the elegance and majesty of the Creator. What were these discoveries all about, and how did they change the direction of science? These are some very big ideas that require a lot of time to think about. Before getting started with the science, let's spend some time with our Bibles.

**→ READ, NARRATE, & DISCUSS**
Genesis 1-2

• What is the same in Genesis 1 and 2, and what is different? How do these two chapters work together to tell us about God?

★ TEACHER NOTE
This lesson consists of reflection on the Christian story of creation. Teachers might choose to allow students to do/review this independently, to engage in the reading/discussion, or to skip this one.

**WEEK 1**

### ⬚ 30m General Science: Grade 6 - Lesson 2

*Making Sense of What We See*

Materials: Plate Tectonics

**→ INTRO**
Some of the discoveries that made scientists think that maybe Things changed over time came from studying rocks. Look at the title and cover. Do you have ideas about plate tectonics? We'll begin the story today with some of those early scientists.

Notice the sidebars in the text, like the one on p.14-15. As you become more independent with your reading, it can be helpful to think about how to deal with these kinds of "rabbit trails," which can enhance the reading. Because it interrupts the main line of thought, it is usually best to finish the section and then come back to the sidebar.

**→ READ & NARRATE**
Ch.1 p.11-13, 16-17 "From some" - "to be connected."

**→ READ, NARRATE, & DISCUSS**
p.14-15 "Around the" - "region's geology."

**WEEK 1**

### ⬚ 30m Labs: Grade 6 - Lesson 1

*Lab: Reading Rocks*

Materials: Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete day 1 of Reading Rocks, as directed in the Lab Book.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

# Science: Grade 6

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## Term 1

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend remaining time gathering materials for next week.

WEEK 2

⬚ **15m Natural History: Grade 6 - Lesson 2**
*An Opportunity*

Materials: Tracking Trash

PREP: Read Teacher Tip.

**→ RECAP**
Recall what this scientist is studying and why. Let's find out how he started tracking trash.

**→ READ & NARRATE**
p.4-7 "Curt didn't" - "Pacific Ocean."

**→ READ, NARRATE, & DISCUSS**
p.8-9 "Finding a certain" - "of the spill."

★ TEACHER TIP
This passage has two parts with two different purposes. The first part tells how this experiment began and is more likely to be narrated as a series of events. The second explains latitude and longitude (and why they are important) and is more likely to be narrated as points that describe these concepts. It can be helpful to teach students to identify the purpose of different passages as they prepare to narrate.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **30m General Science: Grade 6 - Lesson 3**
*Seismic Disasters*

Materials: Plate Tectonics

**→ RECAP**
Recall what we read in the last passage. Today we'll read about seismic activity in Italy and Japan.

**→ READ & NARRATE**
∞ Article Link: The Destruction of Pompeii

**→ READ, NARRATE, & DISCUSS**
p.17-20 “Pompeii” - “with the natural world.”

WEEK 2

⬚ **30m General Science: Grade 6 - Lesson 4**
*Renaissance Ideas*

Materials: Plate Tectonics

PREP: Read Teacher Tip

**→ RECAP**
Tell me about the occurrences in Italy and Japan. Naturalists continued to observe for a long time, but then our story leaps forward during the Renaissance.

**→ READ, NARRATE, & DISCUSS**
p.20-27 "After the decline" - "centuries earlier."

• Nicolaus Steno's work presented, for the first time, the idea that the Earth and even life itself were dynamic. Have you ever learned something so new and surprising that it seemed to change everything? What did you do with this new information?
• While Steno did abandon his scientific pursuits when he entered religious life, many others then and now pursue scientific endeavors and religious vocations hand-in-hand. Daniello Bartoli, Paolo Casati, and Maria Gaetana Agnesi are just a few from Steno's time period. How many can you find?

**→ SUPPLEMENTAL**
The author notes that ideas often inspire people who come later. One scientist you should know who is not mentioned in our book is Mary Anning:
∞ Video Link: Mary Anning (3:40)

★ TEACHER NOTE
Conflict between religion and Steno's work is alluded to on p.25: "Like many of" - "remainder of his life." Teachers might choose to allow it to pass by the way for now (this idea is a major theme in Grade 7-8), to engage in discussion, or to check out the book about Steno in Extra Helpings. The same conflict is mentioned in the supplemental video (1:16-1:31).

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 2

⬚ **30m Labs: Grade 6 - Lesson 2**
*Lab: Reading Rocks*

Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 2 of Reading Rocks, as directed in the Lab Book.

Suggested day 2: Students complete roughly steps 1-3 of the Procedure today. There is some flexibility to complete more or less, depending on the student.

# Science: Grade 6

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## Term 1

**WEEK 3**

⬚ **15m Natural History: Grade 6 - Lesson 3**
*Drift Experiments*

▯ Materials: Tracking Trash

**→ RECAP**
Remind me how Curt started tracking trash. Scientists have known for a long time that ocean motion isn't random. How do you think scientists typically study ocean currents?

**→ READ, NARRATE, & DISCUSS**
p.11-15 "Throughout this" - "floating sneakers."

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

**WEEK 3**

⬚ **30m General Science: Grade 6 - Lesson 5**
*Rocks and Hutton's Theory*

▯ Materials: Plate Tectonics

**→ RECAP**
Tell me about what Renaissance thinkers brought to the story. Before we read about the theory of plate tectonics, look at the figure on p.28. You may continue to look back at it during the passage.

**→ READ, NARRATE, & DISCUSS**
p.29-35 "We now know" - "parts of the puzzle?"

• Discuss the main points of James Hutton's theory. Use words, pictures, or diagrams.

★ TEACHER NOTE
The age of the Earth is mentioned along with Hutton's Theory of the Earth on p.33-34: "He also claimed" - "wind or rain." Teachers might engage in discussion, present alternative theories, or check one of the books in Extra Helpings for additional discussion.

**• NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

**WEEK 3**

⬚ **30m General Science: Grade 6 - Lesson 6**
*Oceanography and Seismology*

▯ Materials: Plate Tectonics

**→ RECAP**
Recall the last reading. We've read about paleontology and geology. Today we have two more types of study that will reveal more puzzle pieces: oceanography and seismology.

**→ READ & NARRATE**
p.35-36 "Exploring the Oceans" - "sea level."

**→ READ, NARRATE, & DISCUSS**
p.36-41 "Finding ways" - "accurate measurement."

**WEEK 3**

⬚ **30m Labs: Grade 6 - Lesson 3**
*Lab: Reading Rocks*

▯ Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 3 of Reading Rocks, as directed in the Lab Book.

Suggested day 3: Students complete steps 4-8 of the Procedure today.

**• NATURE NOTEBOOK**
Note that step 8 of lab directs students to conduct outdoor work before the next lab period: see prompt in Outdoor Work Quick Link.

**WEEK 4**

⬚ **15m Natural History: Grade 6 - Lesson 4**
*Modeling Ocean Motion*

▯ Materials: Tracking Trash

PREP: Read Teacher Tip.

**→ RECAP**
How were the scientists studying the ocean in the last passage? Today, we'll find out what happened when they used the OSCURS program to study the sneaker spill. We'll also learn about different types of ocean motion.

**→ READ & NARRATE**
p.15-17 "In 1992" - "more recoveries."

**→ READ, NARRATE, & DISCUSS**
p.18-19 "Imagine you" - "Indian Ocean."

**→ SUPPLEMENTAL**
∞ Video Link: How Do Ocean Currents Work? (4:34)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, liven up the lesson, or add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Science: Grade 6

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## Term 1

WEEK 4

**⬚ 30m General Science: Grade 6 - Lesson 7**
*Continental Drift*

▯ Materials: Plate Tectonics

**➔ RECAP**
Recall the last passage. Today we begin to put the puzzle together.

**➔ READ, NARRATE, & DISCUSS**
p.41, 43-46 "Continental Drift" - "and accepted."

• Discuss the main points of Alfred Wegener's theory. Use words, pictures, or diagrams.
• Do you think the idea of continental drift impacted the way that scientists organize and classify living things? How might it influence their understanding of which plants and animals are more closely related?

**➔ SUPPLEMENTAL**
∞ Resource Link: NatGeo Continental Drift

**➔ NOTE**
Read Abraham Ortelius p.42 for interest, if desired.

★ TEACHER NOTE
The age of the Earth is mentioned along with Wegener's theory of continental drift on p.43-45 and in the supplemental link: "Wegener believed" - "and Canada." Teachers might engage in discussion or consider alternative theories they would like to share.

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WEEK 4

**⬚ 30m General Science: Grade 6 - Lesson 8**
*The Theory is Born*

▯ Materials: Plate Tectonics

**➔ RECAP**
Recall the last reading. Today we'll learn a bit about Alfred Wegener.

**➔ READ, NARRATE, & DISCUSS**
p.49-52 "The theory" - "moon and Mars."

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WEEK 4

**⬚ 30m Labs: Grade 6 - Lesson 4**
*Lab: Reading Rocks*

▯ Materials: Grade 6 Lab Book and materials listed within

**➔ LAB DAY**
Complete day 4 of Reading Rocks, as directed in the Lab Book.

Suggested day 4: Students complete step 9 of the Procedure today, as well as the Analysis and Conclusion section.

---

WEEK 5

**⬚ 15m Natural History: Grade 6 - Lesson 5**
*A Variation*

▯ Materials: Tracking Trash

PREP: Read Teacher Tip

**➔ RECAP**
What did the scientists learn from the OSCURS computer program? What other kinds of objects do you think would make a good study? Look at the picture on p.20.

**➔ READ, NARRATE, & DISCUSS**
p.21-26 "While they" - "brand of oceanography."

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the ocean or ocean currents? More interest/ease in thinking about how the ocean impacts climate or wildlife? If not, consider observing together more often, exploring extra helpings, or taking a field trip to the ocean or waterway.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 5

**⬚ 30m General Science: Grade 6 - Lesson 9**
*Convection and Plate Tectonics*

▯ Materials: Plate Tectonics

**➔ RECAP**
Recall the last reading. Today we'll learn a bit about two more scientists.

**➔ READ & NARRATE**
p.52-55, 58 "Dan McKenzie" - "discussed below."

**➔ READ, NARRATE, & DISCUSS**
p.56-57 "Jason Morgan" - "plate tectonics."

# Science: Grade 6

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## Term 1

WEEK 5

⬚ **30m General Science: Grade 6 - Lesson 10**
*Measuring Earthquakes*

Materials: Plate Tectonics

PREP: Read Teacher Tip

**→ RECAP**
Recall the last reading. Our focus today is on the two scientists who developed the Richter scale, which is used to measure earthquake strength.

**→ READ, NARRATE, & DISCUSS**
p.58-63 “Charles Richter” - “Uppsala in 1955.”

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in volcanoes, earthquakes, or other aspects of plate tectonics? More interest/ease in discussing how scientists developed the theory over time? If not, consider discussing together more often, exploring extra helpings, or taking a field trip to a natural history museum. Reach out through Contact Us, if you need help.

WEEK 5

⬚ **30m Labs: Grade 6 - Lesson 5**
*Lab: Fossils*

Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete Fossils, as directed in the Lab Book.

WEEK 6

⬚ **15m Natural History: Grade 6 - Lesson 6**
*Riding the Currents*

Materials: Tracking Trash

**→ RECAP**
Recall what happened in the last passage. These ocean scientists continue to work with citizen scientists to make some interesting discoveries. So, we'll read about that and some microscopic ocean creatures that depend on ocean motion today.

**→ READ & NARRATE**
p.26-28 "It is at" - "to track it."

**→ READ, NARRATE, & DISCUSS**
p.29-31 "Sneakers" - "all of us."

**→ SUPPLEMENTAL**
∞ Video Link: The Story of Plankton (3:01)
∞ Image Link: What is Upwelling?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ **30m General Science: Grade 6 - Lesson 11**
*Seafloor Spreading*

Materials: Plate Tectonics

**→ RECAP**
Recall the last passage. Today we read about a scientist who studied the ocean floor.

**→ READ, NARRATE, & DISCUSS**
p.63-64 "Harry Hess" - "sister planets."

★ TEACHER NOTE
Evolution is mentioned at the end of the reading on p.64. Teachers might allow it to pass by the way or consider how they would like to discuss.

• **NATURE NOTEBOOK**
Prompt for General Science in Quick Link.

WEEK 6

⬚ **30m General Science: Grade 6 - Lesson 12**
*The Earthquake Zone*

Materials: Plate Tectonics

PREP: Read Teacher Tip

**→ RECAP**
Recall the last passage. Today we read about two scientists who studied the Ring of Fire.

**→ READ, NARRATE, & DISCUSS**
p.64-68 "Kiyoo Wadati" - "in 1968."

**→ SUPPLEMENTAL**
∞ Video Link: Ring of Natural Disasters (6:34)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 6

⬚ **30m Labs: Grade 6 - Lesson 6**
*Lab: What Do Volcanoes Tell Us?*

Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 1 of What Do Volcanoes Tell Us?, as directed in the Lab Book.

# Science: Grade 6

## Term 1

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. Use any remaining time to begin the Procedure.

**WEEK 7**

### ⬚ 15m Natural History: Grade 6 - Lesson 7

*Where Do Floating Objects Go?*

Materials: Tracking Trash

**→ RECAP**
Recall what we read last week. They seem to use the OSCURS program a lot. How do you think a computer simulation is helpful if it's just a hypothetical experiment?

**→ READ**
p.33 "What would happen" - "might look like."

**→ VIEW**
Examine the figure on p.32, following the parts of the figure using the caption. If students want to look back at the North Pacific Subtropical Gyre, refer to p.19.

**→ READ, NARRATE, & DISCUSS**
p.34-37 "At the beginning" - "Garbage Patch."

**→ SUPPLEMENTAL**
∞ Image Link: Garbage Patch Photos

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

**WEEK 7**

### ⬚ 30m General Science: Grade 6 - Lesson 13

*Geological Dating*

Materials: Plate Tectonics

**→ RECAP**
Recall the last passage. Today we read about our last scientist who developed a method of geological dating.

**→ READ, NARRATE, & DISCUSS**
p.68-69 "Arthur Holmes" - "the earth."

**→ SUPPLEMENTAL**
∞ Video Link: How does radiocarbon dating work? (2:11)
∞ Video Link: Earth's Entire History (4:38)

★ TEACHER NOTE
The age of the Earth is mentioned at the top of p.69 and in the second supplemental video. Isotope dating is introduced in the first supplemental video. Teachers might allow it to pass by the way, engage in discussion, consider alternative theories they would like to share, or skip the videos.

**WEEK 7**

### ⬚ 30m General Science: Grade 6 - Lesson 14

*Discovering Puzzle Pieces*

Materials: Plate Tectonics

**→ RECAP**
Recall the last passage. Let's look at how all of this comes together in the theory of plate tectonics.

**→ READ, NARRATE, & DISCUSS**
p.71-76 "In 1912" - "continental drift."

• Scientists are currently debating whether spreading, like Harry Hess and Marie Tharp described, is a requirement for life on a planet and whether Earth is unique in this feature. Why would they theorize that seafloor spreading would be important for life? What kind of evidence do you suppose they might look for to support or refute such a theory?

★ TEACHER NOTE
The timeline of Earth's magnetic reversal is mentioned on p.73: "every three hundred thousand to seven hundred thousand years." Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

**WEEK 7**

### ⬚ 30m Labs: Grade 6 - Lesson 7

*Lab: What Do Volcanoes Tell Us?*

Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 2 of What Do Volcanoes Tell Us?, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today.

**WEEK 8**

### ⬚ 15m Natural History: Grade 6 - Lesson 8

*The Garbage Patch*

Materials: Tracking Trash

**→ RECAP**
Tell me what they found out in the last passage. What do you think happens to everything in the Garbage Patch?

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Science: Grade 6

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## Term 1

→ **READ & NARRATE**
p.37-39 "What happens" - "the problem."

→ **READ, NARRATE, & DISCUSS**
p.40 "People often" - "same thing!"

• Discuss the chain of cause and effect that begins with trash in the environment.

---

**WEEK 8**

⬚ **30m General Science: Grade 6 - Lesson 15**
*How it Works*

Materials: Plate Tectonics

→ **RECAP**
Tell me what we read last time: how is this theory coming together so far? Let's read about the different movements that can occur according to the theory.

→ **READ, NARRATE, & DISCUSS**
p.76-83 "The Breakthrough" - "opposite directions."

★ TEACHER NOTE
The age of the Earth is mentioned on p.78 ("Some three hundred" - "change is continuous."), 79 ("but over thousands" - "amounts of movement."), and 81 ("Millions of years" - "ten million years."). Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

---

**WEEK 8**

⬚ **30m General Science: Grade 6 - Lesson 16**
*Natural Disasters*

Materials: Plate Tectonics

→ **RECAP**
Tell me about the different types of plate movement according to plate tectonic theory. Today we read about the results of that movement: earthquakes, tsunamis, and volcanoes.

→ **READ, NARRATE, & DISCUSS**
p.83-89 "Earthquakes" - "plate tectonics."

• Describe earthquakes and volcanoes in terms of plate tectonic theory. You may use words, pictures, or diagrams.

★ TEACHER NOTE
**The age of the Earth is mentioned on p.88:** "Recent studies suggest" - "the earth's existence." Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

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**WEEK 8**

⬚ **30m Labs: Grade 6 - Lesson 8**
*Lab: What Do Volcanoes Tell Us?*

Materials: Grade 6 Lab Book and materials listed within

→ **LAB DAY**
Complete day 3 of What Do Volcanoes Tell Us?, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.

---

**WEEK 9**

⬚ **15m Natural History: Grade 6 - Lesson 9**
*Ghost Nets*

Materials: Tracking Trash

→ **RECAP**
What did we learn in the last reading? When cleaning your room, have you ever found unexpected stuff that you didn't even know you'd lost? That's similar to what happens in today's reading.

→ **READ, NARRATE, & DISCUSS**
p.43-46 "In 1991" - "drift in them."

• How do you think knowledge about ocean currents can help?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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**WEEK 9**

⬚ **30m General Science: Grade 6 - Lesson 17**
*Technology*

Materials: Plate Tectonics

→ **RECAP**
Recall what you have learned about plate tectonic theory. Have you read about computer modeling being used to study weather and ocean movement? Do you think that computers could help predict tectonic movement, as well?

→ **READ, NARRATE, & DISCUSS**
p.90-91 "Advances" - "geodynamic models."

# Science: Grade 6

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## Term 1

WEEK 9

### ⬚ 30m General Science: Grade 6 - Lesson 18

*Planning for Plate Tectonics*

▯ Materials: Plate Tectonics

**➔ RECAP**
Recall the last passage. Can you think of ways that people's lives are affected by this understanding of plate tectonics?

**➔ READ & NARRATE**
p.93-95, 98-100 "In previous" - "structural damage."

**➔ READ, NARRATE, & DISCUSS**
p.96-97 "To predict" - "to evacuate."

WEEK 9

### ⬚ 30m Labs: Grade 6 - Lesson 9

*Lab: Earthquake Shake*

▯ Materials: Grade 6 Lab Book and materials listed within

**➔ LAB DAY**
Complete day 1 of Earthquake Shake, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and compose the prelab narration in the lab notebook. Use any remaining time to begin the Procedure.

WEEK 10

### ⬚ 15m Natural History: Grade 6 - Lesson 10

*In Search of Garbage*

▯ Materials: Tracking Trash

**➔ RECAP**
Tell me about the discovery from the last reading. What do you think scientists are doing to solve the problem? What do you think the average person can do?

**➔ READ, NARRATE, & DISCUSS**
p.46-51 "Our first" - "your mother!"

• Interestingly, continued research has discovered that coastal creatures are beginning to colonize the Garbage Patch! You can read more about it in the link, as desired. Does the presence of life change how we think about the Garbage Patch? Should it?

**➔ SUPPLEMENTAL**
∞ Article Link: Garbage Patch Home

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

### ⬚ 30m General Science: Grade 6 - Lesson 19

*Planetary Change*

▯ Materials: Plate Tectonics

**➔ RECAP**
Recall what you read in the last passage. Today's reading tells how plate tectonic theory affects how we study the planets.

**➔ READ, NARRATE, & DISCUSS**
p.100-105 "Plate Tectonics" - "might form."

★ TEACHER NOTE
The timescale of Earth's existence is mentioned on p.105: "Such a process" - "ten million years." Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

WEEK 10

### ⬚ 30m General Science: Grade 6 - Lesson 20

*What Lies Ahead*

▯ Materials: Plate Tectonics

**➔ RECAP**
Recall what you read in the last passage.

**➔ READ, NARRATE, & DISCUSS**
p.106-109 "What Lies" - "years to come."

**➔ SUPPLEMENTAL**
∞ Video Link: Alfred Wegener (7:51)

★ TEACHER NOTE
The timescale of Earth's existence, past and present, is mentioned several times on p.106. Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

# Science: Grade 6

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## Term 1

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|---|---|---|
| WEEK 10 | ⬚ **30m Labs: Grade 6 - Lesson 10**<br>*Lab: Earthquake Shake*<br>▯ Materials: Grade 6 Lab Book and materials listed within<br><br>**➔ LAB DAY**<br>Complete day 2 of Earthquake Shake, as directed in the Lab Book.<br><br>Suggested day 2: Students complete the Procedure today. | |
| WEEK 11 | ⬚ **15m Natural History: Grade 6 - Lesson 11**<br>▯ Materials: Tracking Trash<br><br>PREP: Read Teacher Tip<br><br>**➔ CATCH UP**<br>Use this time to catch up on lessons in this or another course, or explore Extra Helpings. | ★ TEACHER TIP<br>When students take exams next week, evaluate what they recall, but more importantly, whether the ocean, its currents, or its connection to the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?<br><br>**• OUTDOOR WORK**<br>Weekly walks and daily outdoor time are vital! Resources in Quick Links. |
| WEEK 11 | ⬚ **30m General Science: Grade 6 - Lesson 21**<br>*Catch Up*<br>▯ Materials: Plate Tectonics<br><br>PREP: Read Teacher Tip<br><br>**➔ CATCH UP**<br>Use this time to catch up on lessons in this or another course, or explore Extra Helpings. | ★ TEACHER TIP<br>When student(s) take exams next week, evaluate what they recall, but more importantly, whether plate tectonics, its effects, or how scientists arrived at the theory are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help. |
| WEEK 11 | ⬚ **30m General Science: Grade 6 - Lesson 22**<br>*Catch Up*<br>▯ Materials: Plate Tectonics<br><br>**➔ CATCH UP**<br>Use this time to catch up on lessons in this or another course, or explore Extra Helpings. | |
| WEEK 11 | ⬚ **30m Labs: Grade 6 - Lesson 11**<br>*Lab: Earthquake Shake*<br>▯ Materials: Grade 6 Lab Book and materials listed within<br><br>**➔ LAB DAY**<br>Complete day 3 of Earthquake Shake, as directed in the Lab Book.<br><br>Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book. | |
| WEEK 12 | ⬚ **15m Natural History: Grade 6 - Lesson 12**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Tracking Trash | **• OUTDOOR WORK**<br>Weekly walks and daily outdoor time are vital! Resources in Quick Links. |
| WEEK 12 | ⬚ **30m General Science: Grade 6 - Lesson 23**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Plate Tectonics | |

# Science: Grade 6

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## Term 1

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|---|---|---|
| **WEEK 12** | ⬚ **30m General Science: Grade 6 - Lesson 24**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Plate Tectonics | |
| **WEEK 12** | ⬚ **30m Labs: Grade 6 - Lesson 12**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Grade 6 Lab Book | |

# Science: Grade 6

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## Term 2

WEEK 1

⬚ **15m Natural History: Grade 6 - Lesson 13**
*Finding Isle Royale*

▯ Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip.

**→ INTRO**
Look at the picture on the cover. Have you seen or read anything about wolves or moose before? If so, what do you know about them? If not, what are your impressions?

**→ READ, NARRATE, & DISCUSS**
p.1-6 "Rising out" - "and explore."

• Have you heard of ecosystems before? How would you describe an ecosystem? (Read the sidebar on p.2, as desired.) Look at the pictures on the page facing p.1 and on p.3. What do these photos tell you about the ecosystem featured in this book?
• Look at the map on p.4. Find Isle Royale on a map of North America. Did you know that there are moose on this tiny island? What else do you think might live there?

★ TEACHER TIP
We won't have time for many of the sidebars in this book, but you can share them with students, as desired.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 1

⬚ **30m General Science: Grade 6 - Lesson 25**
*Horrible Accident in Vermont*

▯ Materials: Phineas Gage

PREP: Read Teacher Tip

**→ INTRO**
This is the story of a man who suffered a terrible head injury. Can you spot it in the picture on the cover?

**→ READ, NARRATE, & DISCUSS**
p.1-8 "The most unlucky" - "happened to him."

•Throughout this course, we'll be learning about anatomy, the study of the body's structures, and physiology, the study of how they work. We organize all of the different parts into systems, even though they really all function together. Why do you think scientists would study it in smaller portions like that? Do you know of any of these systems yet?

★ TEACHER TIP
Note that the next lesson is an activity day!

---

WEEK 1

⬚ **30m General Science: Grade 6 - Lesson 26**
*Body Systems*

▯ Materials: video links, paper, scissors, pencils/markers

PREP: Read Teacher Tip

**→ RECAP**
Recall what happened to Phineas Gage. The skin is part of one system, while the brain is part of another. Do you know what systems they are each part of? Watch the video for an introduction to body systems. Listen for where the skin and brain belong. Were any other systems involved in Phineas's injury?

**→ VIEW, NARRATE, & DISCUSS**
∞ Video Link: Body Systems (3:33)

• Consider how incredible is the formation of our bodies. Consider how many intricate details must go right during the formation of each person. Every living being is truly a miracle!
• An idea that is beautifully displayed in our body systems is that "form follows function." How are the forms in our bodies related to the functions we need?

**→ ACTIVITY**
Let's represent the body systems by making a paper chain, something like in the video link. How many sides do you need to represent the systems? How will you represent each system?
∞ Video Link: Paper Chain People (3:16)

★ TEACHER NOTE
The reproductive system is mentioned in the lesson video (2:17-2:22). Teachers might allow it to pass by the way or decide how they would like to discuss further with students.

★ TEACHER TIP
Note that there is an optional activity next week using the microscope and prepared slides.

---

WEEK 1

⬚ **30m Labs: Grade 6 - Lesson 13**
*Lab: Classification*

▯ Materials: Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete day 1 of Classification, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interests.

★ TEACHER NOTE
Lab 1 introduces the idea of "relatedness" between species. Teachers might choose to allow the idea to pass by the way, to consider how they would like to discuss, or what other theories they might like to present.

# Science: Grade 6

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## Term 2

**WEEK 2**

### ⬚ 15m Natural History: Grade 6 - Lesson 14

*Arriving at Isle Royale*

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip.

**→ RECAP**
What did we read in the last passage? Where are we? Why are we here? Today, we'll begin the story of the place that is Isle Royale.

**→ READ, NARRATE, & DISCUSS**
p.6-13 "The only way" - "their way there."

**→ SUPPLEMENTAL**
∞ Image Link: History and Culture of Isle Royale

★ TEACHER NOTE
The Earth's age is mentioned, and the formation of the land is described on p.9. Teachers might take the opportunity to have a discussion or to share alternative theories.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to liven the lesson, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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**WEEK 2**

### ⬚ 30m General Science: Grade 6 - Lesson 27

*Tending to Phineas*

Materials: Phineas Gage, optional: microscope, prepared slides

PREP: Read Teacher Tip

**→ RECAP**
Recall the last passage. What happened to Phineas? What do you think will happen next?

**→ READ, NARRATE, & DISCUSS**
p.8-15 "That's how Dr." - "postsurgical infections."

• Is this what you expected to happen to Phineas? What most surprised you? What would it have been like if you were one of Phineas's friends on the scene?

**→ SUPPLEMENTAL**
If you have a microscope and prepared slides, this is a great time to compare some of the many different types of cells at work in the body. Compare the dog tissue slides and the blood smear. Describe the different forms/shapes that you see. Consider how the shapes might contribute to different functions.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

★ TEACHER TIP
Note that the next lesson is an additional day to work on Lab 1!

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**WEEK 2**

### ⬚ 30m General Science: Grade 6 - Lesson 28

*Lab: Classification*

Materials: Grade 6 Lab Book and materials listed within

**→ ACTIVITY**
Use this time to continue work on Lab 1.

---

**WEEK 2**

### ⬚ 30m Labs: Grade 6 - Lesson 14

*Lab: Classification*

Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 2 of Classification this week, as directed in the Lab Book.

Suggested day 2: Students work on the Procedure today. This is a long procedure, so a lesson period has been provided for additional time.

---

**WEEK 3**

### ⬚ 15m Natural History: Grade 6 - Lesson 15

*Wildlife at Isle Royale*

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip.

**→ RECAP**
Recall what we learned in the last passage. Let's continue the story of Isle Royale. We'll find out how the moose and wolves connect to the ecosystem and what the author notes when she arrives there.

★ TEACHER TIP
In this passage, the first part introduces us to the dynamics of the ecosystem and the second to the author's first impressions of the ecosystem. Encourage students to recognize these parts and narrate them in turn.

★ TEACHER NOTE
Geologic land formation is referred to on p.19: "...during the last ice sheet retreat." Teachers might allow it to pass by the way

## Term 2

**→ READ, NARRATE, & DISCUSS**
p.13-20 "When the earliest" - "research team."

• We don't have time to read all of the team bios, but look at the pictures of the Petersons on p.21. Lead scientist Rolf Peterson says Isle Royale is an "ideal natural laboratory" because humans have minimal impact here. Trees are not harvested; animals are not hunted or abused. However, Candy Peterson says that this perspective excludes the important role that humans have to play. We should be enjoying, appreciating, and taking responsibility for nature. Labs are for "tinkering," but she says that Isle Royale is more like a giant circus where the animals are in charge and we watch and observe. What do you think about that? As the story continues, think about these perspectives and how people with different perspectives can work together.

or consider alternative theories they might like to share.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

### ⬚ 30m General Science: Grade 6 - Lesson 29

*Wounds*

Materials: video links

**→ RECAP**
Where did we leave Phineas? He's got this terrible wound right now. Let's find out more about what doctors knew and didn't know in 1848, and what needs to happen for his wounds to heal. Students don't need to remember the scientific terms in the video. Focus on the meaning of the ideas.

**→ VIEW, NARRATE, & DISCUSS**
∞ Video Link: Germ Theory (4:39)

• Did you notice how Koch learned about and tested his ideas about cholera? He used what scientists call an animal model, a mouse. The model stands in for a human so that scientists can learn more about how the human body works without harming a human. What do you think are the pros and cons of this? (e.g., good that humans aren't harmed, but not fair to animals; good if the model fits, but bad if we don't realize its limitations; etc.)
• If doctors didn't yet know about germs in 1848, how was Phineas going to heal?

**→ VIEW, NARRATE, & DISCUSS**
∞ Video Link: How Wounds Heal (4:01)

• Which body systems did you notice working together to heal the wound in the video?

WEEK 3

### ⬚ 30m General Science: Grade 6 - Lesson 30

*Phineas Lives*

Materials: Phineas Gage

**→ RECAP**
Remind me where we left Phineas. Let's find out what happens next.

**→ READ, NARRATE, & DISCUSS**
p.15-22 "None of this" - "make arrangements."

• Dr. Harlow said, "I dressed him. God healed him." Why did he say this? What does this tell us about Dr. Harlow?
• Is Phineas healed? What does it mean to be healed? Is healing the same as being cured and exactly as you were before? What happened when Jesus healed? Is this the same or different from what is meant by Phineas's doctors?

WEEK 3

### ⬚ 30m Labs: Grade 6 - Lesson 15

*Lab: Classification*

Materials: Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete day 3 of Classification, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.

★ TEACHER TIP
Students will need fresh, unblemished apples for the next lab!

WEEK 4

### ⬚ 15m Natural History: Grade 6 - Lesson 16

*An Island Out of Balance*

Materials: The Wolves and Moose of Isle Royale

**→ RECAP**
Recall what we learned in the last passage. What do you think are the differences between studying in the laboratory and studying in the field? Let's find out what's going on in this field study.

**→ READ, NARRATE, & DISCUSS**
p.27-31 "The Isle Royale" - "National Park Service."

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## Term 2

• When we study in the laboratory, we usually try to simplify and focus on just one thing, but the field is more complicated. What kinds of things do you see in the pictures and hear in the words of the story that make studying in the field more complicated? Can you understand why Candy Peterson thinks Isle Royale is more like a circus than a laboratory? What do you like or not like about her comparison?
• Discuss the food chain/web based on the reading OR for predators and prey in your area. Draw or diagram, if helpful.

---

WEEK 4

### ⬚ 30m General Science: Grade 6 - Lesson 31

*How We Know What We Know*

▯ Materials: video links

**→ RECAP**
Phineas Gage lives, but he still has a problem. Tell me about that. Part of his problem is that doctors know so little about how the brain works. Even for basic, everyday tasks, our bodies are performing an amazing dance. Let's watch the video for just one example.

**→ VIEW, NARRATE, & DISCUSS**
∞ Video Link: Oxygen's Surprisingly Complex Journey (5:10)

• How many systems were working together in the video? How do you think we figured all of that out? Let's find out how William Harvey figured out that blood circulates.

**→ VIEW, NARRATE, & DISCUSS**
∞ Video Link: Ceaseless Motion (to 4:54)

• Did you notice how William Harvey finally figured it out? (He used models and performed experiments.)
• Phineas is what we call a case study, a unique opportunity to learn through an example. Scientists will learn a lot about the brain by observing him, but they are limited. Discuss why they are limited. (e.g., they cannot experiment, they cannot understand the complexity of what has happened to him) How do you suppose models help to bring understanding?

---

WEEK 4

### ⬚ 30m General Science: Grade 6 - Lesson 32

*The Doctors*

▯ Materials: Phineas Gage, image link

PREP: Read Teacher Tip

**→ RECAP**
Phineas still has a problem, but there are a lot of doctors who are interested in learning more about his case. Let's do a little picture study to get an idea of what these doctors are like. What do you notice in The Gross Clinic?
∞ Image Link: The Gross Clinic

**→ READ, NARRATE, & DISCUSS**
p.23-27 "In the winter" - "wouldn't be human."

• "...both groups are slightly right, but mostly wrong. Yet their wrong theories... will steer our knowledge of the brain in the right direction." How can someone be both right and wrong? Have you ever experienced this? How can a wrong idea help us get to the truth?

★ TEACHER NOTE
The painting included for picture study is a graphic representation of surgery. Teachers should preview if they have concerns. They might choose to present the picture like any other, to direct students to the setting and doctors rather than the patient, or to skip this introduction.

★ TEACHER TIP
Note that there is an optional activity next week using the microscope and prepared slides.

---

WEEK 4

### ⬚ 30m Labs: Grade 6 - Lesson 16

*Lab: Modeling the Skin*

▯ Materials: Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete day 1 of Modeling the Skin, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. Try to complete step 1 of the Procedure today.

★ TEACHER TIP
If needed, be sure to pour your agar plates according to the package instructions at the appropriate time. They are needed in step 2 of the Procedure.
∞ **Video Link:** Pouring Agar Plates (2:51)

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WEEK 5

### ⬚ 15m Natural History: Grade 6 - Lesson 17

*A Plan to Restore Balance*

▯ Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip

**→ RECAP**
Recall what we learned in the last passage. When ideas and situations become complicated and people have different perspectives, they often disagree about what decisions are best. This happened with Isle Royale, and we'll find out more about that today.

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in wolves or predator-prey relationships? More interest/ease in discussing field work or tough decisions in field work? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve.

**• OUTDOOR WORK**

# Science: Grade 6

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## Term 2

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| | **→ READ, NARRATE, & DISCUSS**<br>p.33-39 "What were" - "four-year-old female."<br><br>**→ SUPPLEMENTAL**<br>Scientists decided that wolf reintroduction was the best decision. Watch this PBS video link to see what happened when this keystone species was reintroduced at Yellowstone.<br>∞ Video Link: Wolves of Yellowstone (5:19) | Weekly walks and daily outdoor time are vital! Resources in Quick Links. |
| **WEEK 5** | ⬚ **30m General Science: Grade 6 - Lesson 33**<br>*The Brain*<br>▯ Materials: Phineas Gage<br><br>PREP: Read Teacher Tip<br><br>**→ RECAP**<br>Recall the last reading.<br><br>**→ READ, NARRATE, & DISCUSS**<br>p.27-34 "The cortex is" - "acts, and reacts."<br><br>• If you have a microscope and prepared slides, check out the pig motor neuron. How does this compare to the other cells you observed previously? How is its form similar/different from the others? How is its function similar/different based on what you have read? | ★ TEACHER TIP<br>Note that the next 3 lessons are a clay modeling activity! |
| **WEEK 5** | ⬚ **30m General Science: Grade 6 - Lesson 34**<br>*Clay Modeling*<br>▯ Materials: clay or other modeling material, toothpicks, knives, or any tools you like for modeling, optional: paint<br><br>PREP: Read Teacher Tip<br><br>**→ INTRO**<br>In the last reading, you read about the form of the brain. Using what you know and the illustrations in your book, create a clay model of the brain over the next 3 lessons.<br><br>**→ ACTIVITY** | ★ TEACHER TIP<br>Are your learner(s) demonstrating more familiarity/interest in the human body or how it works? More interest/ease in discussing how scientists developed their understanding? If not, consider discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a human body exhibit. Reach out through Contact Us, if you need help. |
| **WEEK 5** | ⬚ **30m Labs: Grade 6 - Lesson 17**<br>*Lab: Modeling the Skin*<br>▯ Materials: Grade 6 Lab Book and materials listed within<br><br>**→ LAB DAY**<br>Complete day 2 of Modeling the Skin, as directed in the Lab Book.<br><br>Suggested day 2: Students complete the Procedure today. | |
| **WEEK 6** | ⬚ **15m Natural History: Grade 6 - Lesson 18**<br>*Releasing Wolves*<br>▯ Materials: The Wolves and Moose of Isle Royale<br><br>**→ RECAP**<br>Remind me what the problem is at Isle Royale. Recall what scientists have decided to do about this. How do you think they actually did this?<br><br>**→ READ, NARRATE, & DISCUSS**<br>p.41-47 "The team" - "in all of this."<br><br>• Explain the process of reintroducing a wolf at Isle Royale. What do you think is the most challenging part?<br><br>**→ SUPPLEMENTAL**<br>There are two wolf species featured in our book, which you can read about on p.40-41, if desired. Watch a brief video of the first female wolf being released onto the island. What do you think as you watch her?<br>∞ Video Link: Isle Royale Releasing Wolves (1:15) | **• OUTDOOR WORK**<br>Weekly walks and daily outdoor time are vital! Resources in Quick Links. |
| **WEEK 6** | ⬚ **30m General Science: Grade 6 - Lesson 35**<br>*Clay Modeling*<br>▯ Materials: clay or other modeling material, toothpicks, knives, or any tools you like for modeling, optional: paint<br><br>**→ ACTIVITY** | |

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## Term 2

Continue your model of the brain today.

WEEK 6

⬚ **30m General Science: Grade 6 - Lesson 36**
*Clay Modeling*

> Materials: clay or other modeling material, toothpicks, knives, or any tools you like for modeling, optional: paint

**→ ACTIVITY**
Complete your model of the brain today.

WEEK 6

⬚ **30m Labs: Grade 6 - Lesson 18**
*Lab: Modeling the Skin*

> Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 3 of Modeling the Skin, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.

WEEK 7

⬚ **15m Natural History: Grade 6 - Lesson 19**
*Learning from Scat and Fur*

> Materials: The Wolves and Moose of Isle Royale

**→ RECAP**
Tell me about what you remember from the last reading. Once the wolves are released, how do you think the scientists continue to study them?

**→ READ, NARRATE, & DISCUSS**
p.51-58 "Morgan and I" - "to the lodge."

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ **30m General Science: Grade 6 - Lesson 37**
*The Brain Debate*

> Materials: Phineas Gage

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.34-42 "The Boston Doctors" - "out of town."

• Discuss the arguments of the Whole-Brainers and the Localizers. Use an outline, if helpful.

**→ SUPPLEMENTAL**
∞ Video Link: The Great Brain Debate (5:20)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 7

⬚ **30m General Science: Grade 6 - Lesson 38**
*Following Phineas*

> Materials: Phineas Gage

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.43-49 "The story of" - "everywhere Phineas goes."

WEEK 7

⬚ **30m Labs: Grade 6 - Lesson 19**
*Lab: Modeling the Heart*

> Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 1 of Modeling the Heart, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction.

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## Term 2

**WEEK 8**

### ⬚ 15m Natural History: Grade 6 - Lesson 20

*Learning from Moose*

Materials: The Wolves and Moose of Isle Royale

**→ RECAP**
Remind me what the last reading was about. Let's find out how scientists learn from the moose today.

**→ READ, NARRATE, & DISCUSS**
p.58-66 "We hike miles" - "research firsthand."

**→ SUPPLEMENTAL**
∞ Video Link: Outside Science (4:34)

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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**WEEK 8**

### ⬚ 30m General Science: Grade 6 - Lesson 39

*The end of Phineas*

Materials: Phineas Gage

**→ RECAP**
Recall the last reading.

**→ READ, NARRATE, & DISCUSS**
p.49-57 "In 1859, Phineas" - "shipped to Massachusetts?"

• Let's return to this question of whether Phineas is healed. Does he seem healed? Is he trying to heal? What do you think when you hear about his life?

---

**WEEK 8**

### ⬚ 30m General Science: Grade 6 - Lesson 40

*Doctors Again*

Materials: Phineas Gage

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.57-64 "What a request." - "stop acting human?"

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**WEEK 8**

### ⬚ 30m Labs: Grade 6 - Lesson 20

*Lab: Modeling the Heart*

Materials: Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 2 of Modeling the Heart, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure and the Analysis and Conclusions today.

---

**WEEK 9**

### ⬚ 15m Natural History: Grade 6 - Lesson 21

*Finding Moose*

Materials: The Wolves and Moose of Isle Royale

**→ RECAP**
Tell me how the wolves and moose are studied at Isle Royale. Do you think the scientists get to watch the real-life animals? How often do you think that happens? What do you think that would be like?

**→ READ, NARRATE, & DISCUSS**
p.69-74 "We spot" - "us to sleep."

• What are some of the activities of a field scientist while outdoors? Compare/contrast to what you do when you are on a walk.
• Make up a poem about wolves together!

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

---

**WEEK 9**

### ⬚ 30m General Science: Grade 6 - Lesson 41

*Putting Phineas Together Again*

Materials: Phineas Gage

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.65-69 "In our time" - "down to one."

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## Term 2

**➜ SUPPLEMENTAL**
∞ Article & Image Link: Smithsonian Magazine

**WEEK 9**

### ⬚ 30m General Science: Grade 6 - Lesson 42

*Putting Phineas Together Again*

Materials: Phineas Gage

**➜ RECAP**
Recap the last reading.

**➜ READ, NARRATE, & DISCUSS**
p.69-75 "Brainvox plots it" - "Gage drove fast."

• What makes us human? Is it any of the ideas that the author suggests? Or is it something else? If someone loses these abilities through birth, injury, or illness, do they cease to be human? If your faith tradition provides guidance, what does it say?

★ TEACHER NOTE
The question of what makes us human is introduced on p.70: "Humans have" - "an iron rod." Teachers might choose to allow it to pass by the way or to discuss the philosophical/theological implications.

**WEEK 9**

### ⬚ 30m Labs: Grade 6 - Lesson 21

*Lab: Student Design*

Materials: Grade 6 Lab Book and materials listed within, video and image links

PREP: Read Teacher Tip

**➜ LAB DAY**
Complete day 1 of Student Design, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. They should communicate their plan and any needed supplies to their teacher.

∞ Video Link: Anatomy T-shirt (3:55)
∞ Image Link: Anatomy Cake

★ TEACHER TIP
Teachers should verify that students have what they need to begin their model next week.

**WEEK 10**

### ⬚ 15m Natural History: Grade 6 - Lesson 22

*Visiting the Circus*

Materials: The Wolves and Moose of Isle Royale

**➜ RECAP**
Tell me what you have learned about the wolves and moose at Isle Royale. Let's finish the story and find out how the ecosystem is today.

**➜ READ, NARRATE, & DISCUSS**
p.76-82 "Let's return" - "and his DNA."

• Compare Candy's perspective to man's appointed task of dominion over creation (Genesis 1:28).

**➜ SUPPLEMENTAL**
∞ Video Link: Isle Royale Nature Video (4:08)

★ TEACHER NOTE
**Evolution is mentioned on p.79:** "... have evolved into..." Teachers might choose to let it pass by the way or discuss with students.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

**WEEK 10**

### ⬚ 30m General Science: Grade 6 - Lesson 43

*An Update on Phineas in Chile*

Materials: article link

PREP: Read Teacher Tip

**➜ RECAP**
Recap the last reading.

**➜ READ, NARRATE, & DISCUSS**
∞ Article Link: The Curious Case of Phineas Gage

• Does this portrayal of Phineas in Chile affect your ideas about his life? Why or why not?

★ TEACHER TIP
Note that the next lesson is an additional day to work on the final lab project!

**WEEK 10**

### ⬚ 30m General Science: Grade 6 - Lesson 44

*Lab: Student Design*

Materials: Grade 6 Lab Book and materials listed within

**➜ ACTIVITY**
Use this time to continue work on the final model project.

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## Term 2

→ **SUPPLEMENTAL**
A fun video to watch while you think about your amazing body!
∞ Video Link: How Old is Your Body? (3:09)

**WEEK 10**

⬚ **30m Labs: Grade 6 - Lesson 22**
*Lab: Student Design*

Materials: Grade 6 Lab Book and materials listed within

→ **LAB DAY**
Complete day 2 of Student Design, as directed in the Lab Book.

Suggested day 2: Students will work on their model for the remainder of the term, completing the Analysis and Conclusions when finished. To allow them plenty of time, a lesson period has been provided for additional time.

**WEEK 11**

⬚ **15m Natural History: Grade 6 - Lesson 23**

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip

→ **CATCH UP**
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether wolves, the role of predators in an ecosystem, or field science are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

**WEEK 11**

⬚ **30m General Science: Grade 6 - Lesson 45**
*Catch Up*

Materials: Phineas Gage

PREP: Read Teacher Tip

→ **CATCH UP**
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether anatomy, physiology, or how scientists arrived at their understanding are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.

**WEEK 11**

⬚ **30m General Science: Grade 6 - Lesson 46**
*Catch Up*

Materials: Phineas Gage

→ **CATCH UP**
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

**WEEK 11**

⬚ **30m Labs: Grade 6 - Lesson 23**
*Lab: Student Design*

Materials: Grade 6 Lab Book and materials listed within

→ **LAB DAY**
Complete day 3 of Student Design, as directed in the Lab Book.

Suggested day 3: Students complete their model and finish their Analysis and Conclusions.

**WEEK 12**

⬚ **15m Natural History: Grade 6 - Lesson 24**
*Exam Week*

→ **EXAMS**
Answer question(s) related to course.

Questions will come from:
The Wolves and Moose of Isle Royale

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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## Term 2

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| WEEK 12 | ⬚ **30m General Science: Grade 6 - Lesson 47**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Phineas Gage | |
| WEEK 12 | ⬚ **30m General Science: Grade 6 - Lesson 48**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Phineas Gage | |
| WEEK 12 | ⬚ **30m Labs: Grade 6 - Lesson 24**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Grade 6 Lab Book | |

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## Term 3

WEEK 1

### ⬚ 15m Natural History: Grade 6 - Lesson 25

*Frogging*

Materials: The Frog Scientist

PREP: Read Teacher Tip.

**→ INTRO**
Why do you think frogs are important? What can we learn from them? Do they impact humans? This is a story about scientists who study how frogs grow in an environment contaminated with pesticides.

**→ READ, NARRATE, & DISCUSS**
p.1-3 "The sun is" - "through the mud."

• Have you ever caught or raised tadpoles? Would you like to? How do they feel in your hand, or how do you imagine they would feel in your hand?

**→ SUPPLEMENTAL**
∞ Video Link: Do We Really Need Pesticides? (5:18)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, liven the lesson, or add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

### ⬚ 30m General Science: Grade 6 - Lesson 49

*From Cave to Skyscraper*

Materials: Art of Construction

PREP: Read Teacher Tip

**→ INTRO**
Look at the title and cover. In this book, you will learn how man has used the forces of nature to develop knowledge in structural design. From primitive tents to modern pneumatic roofs, architecture may be one of man's oldest scientific pursuits. Note that you will need your text for lab this term because the lab instruction comes from the text.

**→ READ, NARRATE, & DISCUSS**
p.1-8 "Thirty thousand" - "and pins."

**→ SUPPLEMENTAL**
∞ Image Link: Former Sears Tower
∞ Image Link: Burj Khalifa (tallest building)

★ TEACHER NOTE
Reading begins with a mention of the age of the Earth: "Thirty thousand years..." Teachers might choose to allow it to pass by the way, decide how to discuss, or consider alternative theories they would like to present.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 1

### ⬚ 30m General Science: Grade 6 - Lesson 50

*Building a Tent*

Materials: Art of Construction, lab notebook

**→ RECAP**
Recap the last reading. Today is day 1 (the Introduction) of the Building a Tent lab, so that students will be ready for a full day of activity on lab day.

**→ READ, NARRATE, & DISCUSS**
p.9-10 "The purpose of" - "makes it stand up."

**→ LAB NOTEBOOK**
In your lab notebook, compose a few sentences to answer the introductory questions, "What do I know about building a tent?" and "What do I plan to do?" If you have extra time, copy some of the diagrams from your book.

WEEK 1

### ⬚ 30m Labs: Grade 6 - Lesson 25

*Lab: Building a Tent*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete day 2 of Building a Tent, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interests.

★ TEACHER TIP
Suggested day 1 for the lab Building a Tent was scheduled in Lesson 50, so that students would be prepared for a full lesson of activity on lab day. If students have not yet completed Lesson 50, then teachers may want to complete day 1 together before beginning the lab activity.

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## Term 3

WEEK 2

### ⬚ 15m Natural History: Grade 6 - Lesson 26

*...And Working*

Materials: The Frog Scientist

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.4-7 "Forty-eight..." - "in our water?"

• Is it okay to sacrifice some animals to save others or to save human beings? Try to understand both points of view.

★ TEACHER NOTE
Scientists cull frogs, and the feminizing effect of chemicals on some frogs is described on p.6. Teachers might choose to let them pass by the way or to discuss one or both topics with students.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

### ⬚ 30m General Science: Grade 6 - Lesson 51

*Building a Tent*

Materials: Art of Construction, lab notebook

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading. Today is day 3 (the Analysis and Conclusions) of the Building a Tent lab, so students will continue their reading.

**→ READ, NARRATE, & DISCUSS**
p.11-15 "The straw" - "from spreading apart."

**→ LAB NOTEBOOK**
In your lab notebook, compose a few sentences to answer the questions, "What did I learn?" and "What would I do differently if I built this structure again?" You may want to include a picture or diagram of your tent.

★ TEACHER TIP
Three readings are scheduled this week rather than having an activity during lab. Teachers wanting to continue Lab 1 or do a supplemental activity during the lab time will need to adjust the schedule or have students catch up on the reading at another time.

WEEK 2

### ⬚ 30m General Science: Grade 6 - Lesson 52

*What is a Beam?*

Materials: Art of Construction

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.17-20 "Some elements" - "in the beams."

• Describe tension and compression using words, drawings, and/or diagrams.

**→ NOTE**
The next reading (p.21-26 "Since all" - "bronze rods.") occurs in this week's lab, so be sure to read it before continuing to the next lesson.

WEEK 2

### ⬚ 30m Labs: Grade 6 - Lesson 26

*Lab: Tension and Compression*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete Tension and Compression, as directed in the Lab Book.

WEEK 3

### ⬚ 15m Natural History: Grade 6 - Lesson 27

*The Frog Scientist*

Materials: The Frog Scientist

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.8-11 "Tyrone didn't worry" - "frogs were dying."

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

### ⬚ 30m General Science: Grade 6 - Lesson 53

*Behavior of Materials*

Materials: Art of Construction

**→ RECAP**
Recap the last reading.

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## Term 3

**→ READ, NARRATE, & DISCUSS**
p.26-31 "To get" - "collapsing entirely."

**→ SUPPLEMENTAL**
The Tacoma Narrows Bridge was nicknamed "Galloping Gertie" by construction workers. The rhythmic movement increased over time until its collapse (as seen in the video link) just 4 months later. The footage demonstrates the impressive strength and elasticity of the materials, even though they could not withstand the forces of nature.
∞ Video Link: Tacoma Narrows Bridge (2:35)

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WEEK 3

### ⬚ 30m General Science: Grade 6 - Lesson 54

*The Floor of Your Room*

Materials: Art of Construction

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.33-37 "If the earth" - "design engineer."

• Try calculating the total load on the floor of your room!

**→ NOTE**
The next reading (p.39-41 "The best way" - "near the column axis.") occurs in this week's lab, so be sure to read it before continuing to the next lesson.

**• NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

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WEEK 3

### ⬚ 30m Labs: Grade 6 - Lesson 27

*Lab: A Steel Frame... Made Out of Paper*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 1 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and step 1 of the Procedure.

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WEEK 4

### ⬚ 15m Natural History: Grade 6 - Lesson 28

*Fragile Frogs*

Materials: The Frog Scientist

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.12-17 "Amphibian scientists" - "world's amphibians?"

**→ SUPPLEMENTAL**
∞ Resource Link: Keeping Tadpoles

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 4

### ⬚ 30m General Science: Grade 6 - Lesson 55

*Constructing for Strength*

Materials: Art of Construction, ruler

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.41-46 "Since I-columns" - "(Figure 6.14)."

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WEEK 4

### ⬚ 30m General Science: Grade 6 - Lesson 56

*A Steel Frame... Made Out of Paper*

Materials: Art of Construction

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.47-48 "In your paper" - "and their connections."
Be sure that you understand the structure you will build. Use your lesson time today to think about how you will use the columns/beams previously made. How will you assemble your structure? If helpful, you may want to outline or diagram your plans in your lab notebook, so you have a clear idea for the next lesson.

**• NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

# Science: Grade 6

## Term 3

WEEK 4

### ⬚ 30m Labs: Grade 6 - Lesson 28

*Lab: A Steel Frame... Made Out of Paper*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 2 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure. Help them learn to plan ahead, allowing glue to dry between sessions.

WEEK 5

### ⬚ 15m Natural History: Grade 6 - Lesson 29

*Pesticides*

Materials: The Frog Scientist

PREP: Read Teacher Tip

**→ RECAP**
Recall the last reading.

**→ READ, NARRATE, & DISCUSS**
p.17-19 "The answer" - "to atrazine."

• Consider alternatives to pesticides. Maybe check out this video:
∞ Video Link: Hungry Ducks (2:30)

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in frogs, their life cycle, or their sensitivity to ecological factors? More interest/ease in discussing field work or lab experiments? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

### ⬚ 30m General Science: Grade 6 - Lesson 57

*The Foundation*

Materials: Art of Construction

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.51-56 "The construction" - "going on."

WEEK 5

### ⬚ 30m General Science: Grade 6 - Lesson 58

*Forces that Unbalance*

Materials: Art of Construction, optional: book, cereal box, sandpaper, a few rocks, fan/hairdryer, plastic ruler

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.57-63 "The purpose" - "the ruler."

• Explain equilibrium using words, drawings, and/or diagrams.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in structural design or engineering? More interest/ease in discussing the art of engineering? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a science museum with engineering or building exhibits. Reach out through Contact Us, if you need help.

WEEK 5

### ⬚ 30m Labs: Grade 6 - Lesson 29

*Lab: A Steel Frame... Made Out of Paper*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 3 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 3: Students continue the Procedure.

WEEK 6

### ⬚ 15m Natural History: Grade 6 - Lesson 30

*The Amphibian Ark*

Materials: The Frog Scientist

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.20-21 "In June 2005," - "Jump in!"

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Science: Grade 6

## Term 3

**→ SUPPLEMENTAL**
Have you ever seen frogs from other places in the world? If so, recall one. If not, explore this link:
∞ Resource Link: Amphibia Web

**WEEK 6**

### ⬚ 30m General Science: Grade 6 - Lesson 59

*Wind Drift*

Materials: Art of Construction, optional: fan/hairdryer, needle, thread, cereal box, weight (e.g., spool or metal washer)

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.65-69 "The horizontal" - "weight substantially."

• Calculate the force of wind on the building where you live or school as described on p.65. What would be the wind force of a 50 mph wind? A 100 mph wind?

**→ SUPPLEMENTAL**
∞ Video Link: John Hancock Tower (1:17)
∞ Video Link: John Hancock Tower Update (1:06)

★ TEACHER TIP
There are many constructions in the book that we don't have time for in the lesson schedule, but students can try any of these based on interest and as time permits!

**WEEK 6**

### ⬚ 30m General Science: Grade 6 - Lesson 60

*Earthquakes*

Materials: Art of Construction, optional: sandpaper, cereal box, hammer, nail, block of scrap wood, rope

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.70-74 "The tuned" - "people died."

**→ SUPPLEMENTAL**
∞ Video Link: Swaying Skyscrapers (4:15)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

**WEEK 6**

### ⬚ 30m Labs: Grade 6 - Lesson 30

*Lab: A Steel Frame... Made Out of Paper*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 4 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 4: Students continue the Procedure.

**WEEK 7**

### ⬚ 15m Natural History: Grade 6 - Lesson 31

*From Lab to Field*

Materials: The Frog Scientist

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.22-27 "Young remembers her" - "harming frogs."

• If you have raised frogs, describe how they develop from egg to frog. If not, check out this link:
∞ Video Link: Raising frogs - Frog development (6:57)

★ TEACHER NOTE
The problem of pesticides feminizing the frogs is further described on p.26-27. Teachers might choose to discuss further with students.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

**WEEK 7**

### ⬚ 30m General Science: Grade 6 - Lesson 61

*Using Strings to Carry Load*

Materials: Art of Construction, optional: thread/string, 3-7 paper clips, 3-7 weights (e.g., spools or metal washers)

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.75-80 "Wires and" - "from it."

# Science: Grade 6

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## Term 3

WEEK 7

⬚ **30m General Science: Grade 6 - Lesson 62**
*Suspension Bridges*

Materials: Art of Construction, optional: wooden ruler, thread/string, weight

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.80-83 "The adaptability" - "suspension bridges."

**→ SUPPLEMENTAL**
∞ Image Link: Akashi-Kaikyo Bridge

WEEK 7

⬚ **30m Labs: Grade 6 - Lesson 31**
*Lab: A Steel Frame... Made Out of Paper*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 5 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 5: Students continue the Procedure, if needed, and the Analysis and Conclusions.

WEEK 8

⬚ **15m Natural History: Grade 6 - Lesson 32**
*The Question*

Materials: The Frog Scientist

PREP: Read Teacher Tip.

**→ RECAP**
Recall the last reading.

**→ READ, NARRATE, & DISCUSS**
p.28-33 "It's early on" - "blue-yellow-yellow jug."

• Discuss the design of the experiment. What question are the scientists asking? How are they making sure the test is unbiased?

★ TEACHER TIP
The traditional names for the manipulated and responding variables are independent and dependent. Be sure students understand how they function in the experiment.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **30m General Science: Grade 6 - Lesson 63**
*From Sticks and Stones to Bridges and Buttresses*

Materials: Art of Construction, optional: thread/string, paper clip, weight, nutcracker

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.85-94 "We have" - "them vertically."

**→ SUPPLEMENTAL**
∞ Image Link: Flying Buttress
∞ Image Link: New River Gorge Bridge

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 8

⬚ **30m General Science: Grade 6 - Lesson 64**
*From Strings and Sticks to Trusses*

Materials: Art of Construction, optional: thread/string, nutcracker, weight

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.95-98 "If useful" - "(Figure 12.7)."

**→ NOTE**
The next reading (p.99-103 "To build a" - "efficient trusses are.") occurs in this week's lab, so be sure to read it before continuing to the next lesson.

WEEK 8

⬚ **30m Labs: Grade 6 - Lesson 32**
*Lab: Strings and Sticks*

Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 1 of Strings and Sticks, as directed in the Lab Book.

# Science: Grade 6

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## Term 3

Suggested day 1: Students complete the Introduction and steps 1-2 of the Procedure today.

WEEK 9

**⬚ 15m Natural History: Grade 6 - Lesson 33**
*The Experiment*

▯ Materials: The Frog Scientist

**➔ RECAP**
Recap the last reading.

**➔ READ, NARRATE, & DISCUSS**
p.33-36 "Jasmin picks up" - "a wicked laugh."

• Notice the diagram of Dr. Hayes's experimental design on p.37. Discuss what makes this an excellent test of his hypothesis.

★ TEACHER NOTE
The group's Fifth of July celebration (rather than Fourth of July) is mentioned on p.36. Teachers might choose to discuss further with students.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

**⬚ 30m General Science: Grade 6 - Lesson 65**
*Using Trusses*

▯ Materials: Art of Construction

**➔ RECAP**
Recap the last reading.

**➔ READ, NARRATE, & DISCUSS**
p.103-108 "Figure 12.14" - "out of trusses."

WEEK 9

**⬚ 30m General Science: Grade 6 - Lesson 66**
*Shape and Strength*

▯ Materials: Art of Construction, optional: paper, pencil, 3-5 books

PREP: Read Teacher Tip

**➔ RECAP**
Recap the last reading.

**➔ READ, NARRATE, & DISCUSS**
p.109-118 "If you" - "its material."

★ TEACHER TIP
There is an optional activity next week that requires 4 raw eggs.

WEEK 9

**⬚ 30m Labs: Grade 6 - Lesson 33**
*Lab: Strings and Sticks*

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**➔ LAB DAY**
Complete day 2 of Strings and Sticks, as directed in the Lab Book.

Suggested day 2: Students continue with the Procedure.

WEEK 10

**⬚ 15m Natural History: Grade 6 - Lesson 34**
*The Answer*

▯ Materials: The Frog Scientist

**➔ RECAP**
Recap the last reading.

**➔ READ, NARRATE, & DISCUSS**
p.38-45 "Tyrone's main lab" - "feminized frogs."

★ TEACHER NOTE
A photograph of preserved frog specimens is shown on p.39. Teachers might choose to cover the image or discuss beforehand with students if concerned they will be disturbed.

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

**⬚ 30m General Science: Grade 6 - Lesson 67**
*Getting Creative and Complex*

▯ Materials: Art of Construction

**➔ RECAP**
Recap the last reading.

**➔ READ, NARRATE, & DISCUSS**
p.119-123 "The simple" - "with the roof."

# Science: Grade 6

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## Term 3

WEEK 10

### ⬚ 30m General Science: Grade 6 - Lesson 68

*Creative and Complex*

> Materials: Art of Construction, optional: 4 raw eggs, 8 egg cups, 8 cotton balls, plywood scrap, scrap fabric or bandana

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.123-133 "Domes have" - "four columns."

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WEEK 10

### ⬚ 30m Labs: Grade 6 - Lesson 34

*Lab: Strings and Sticks*

> Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 3 of Strings and Sticks, as directed in the Lab Book.

Suggested day 3: Students continue with the Procedure.

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WEEK 11

### ⬚ 15m Natural History: Grade 6 - Lesson 35

*Values*

> Materials: The Frog Scientist

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.46-51 "When Tyrone" - " and the same."

• It's very important that we think about our choices. Are they consistent with our values? What are the consequences of our choices? It's also important to know that we'll sometimes make mistakes, and we have to work to correct those mistakes. How do we see this in Dr. Hayes's work?

★ TEACHER NOTE
An interaction between the scientist and his father is considered on p.46-51. Teachers might choose to preread, to use the discussion prompt, or to skip this final lesson.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether frogs, their life cycle, or their sensitivity to ecological factors are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

**• OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 11

### ⬚ 30m General Science: Grade 6 - Lesson 69

*Air as a Construction Material?*

> Materials: Art of Construction, optional: cylindrical balloon, cup that fits balloon, small books, thread/string, weight (e.g., spool or metal washer)

PREP: Read Teacher Tip

**→ RECAP**
Recap the last reading.

**→ READ, NARRATE, & DISCUSS**
p.135-143 "If you" - "man lives."

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether the design, engineering, or art of structures are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help or have any concerns about readiness for the next stage in relationship building.

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WEEK 11

### ⬚ 30m General Science: Grade 6 - Lesson 70

*Catch Up*

> Materials: Art of Construction

**→ CATCH UP**
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

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WEEK 11

### ⬚ 30m Labs: Grade 6 - Lesson 35

*Lab: Strings and Sticks*

> Materials: Art of Construction, Grade 6 Lab Book and materials listed within

**→ LAB DAY**
Complete day 4 of Strings and Sticks, as directed in the Lab Book.

# Science: Grade 6

## Term 3

Suggested day 4: Students complete the Procedure and the Analysis and Conclusions.

| | | |
|---|---|---|
| WEEK 12 | ⬚ **15m Natural History: Grade 6 - Lesson 36**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>The Frog Scientist | • **OUTDOOR WORK**<br>Weekly walks and daily outdoor time are vital! Resources in Quick Links. |
| WEEK 12 | ⬚ **30m General Science: Grade 6 - Lesson 71**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Art of Construction | |
| WEEK 12 | ⬚ **30m General Science: Grade 6 - Lesson 72**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Art of Construction | |
| WEEK 12 | ⬚ **30m Labs: Grade 6 - Lesson 36**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course.<br><br>Questions will come from:<br>Art of Construction | |

  

# Science: Grade 6

## Examination

### Term 1

**General Science: Grade 6**

· Plate Tectonics: Draw or diagram 3 types of plate movement and what they cause. OR Explain what is meant by the terms continental drift and seafloor spreading.

· Explain how you did one investigation in science lab this term and what you learned from it.

**Natural History: Grade 6**

· Tracking Trash: Draw or diagram 3 different factors that help create ocean currents; OR explain how ocean currents affect climate and wildlife.

### Term 2

**General Science: Grade 6**

· Phineas Gage: Explain why animals are used as models in science. OR Tell how the life of Phineas Gage has contributed to our understanding of the brain.

· Explain how you did one investigation in science lab this term and what you learned from it.

**Natural History: Grade 6**

· The Wolves and Moose of Isle Royale: Using words or pictures, explain how wolves affect the ecosystem; OR describe 3 ways that scientists study animals in the field.

### Term 3

**General Science: Grade 6**

· The Art of Construction: Explain how a structure's form comes from its needed function. OR The author describes architecture as both art and engineering. Give 3 examples that demonstrate this idea.

· Explain how you did one construction this term and what you learned from it.

**Natural History: Grade 6**

· The Frog Scientist: Diagram the life cycle of a frog; OR give 2 examples of chemical, biological, or environmental conditions that may impact amphibians in the wild.